

Cross-Lingual Low-Resource NLP for Kannada: Improving Named Entity Recognition Using Multilingual Transformer Models and Knowledge Transfer

A Literature Survey, Problem Formulation, and a MuRIL-Oriented Preprocessing Pipeline

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Abstract

Named Entity Recognition (NER) in Dravidian languages such as Kannada is a canonical sequence-labeling benchmark in low-resource natural language processing [1]. Due to rich agglutinative morphology, complex orthography, and the scarcity of gold-standard token-annotated datasets, building high-precision models remains difficult [14, 12]. This report documents the data-preprocessing and formulation stage of a study whose eventual goal is a fine-tuned cross-lingual Transformer classifier (such as Google’s MuRIL or IndicBERT) [6, 5]. Here we (i) formalize the token-level sequence classification problem, (ii) survey the methodological arc of the field, from lexicon and hand-crafted Conditional Random Field (CRF) baselines [8] through distributed representations and deep sequence models (BiLSTM-CRF) [9, 10, 11] to pre-trained multilingual Transformers (mBERT, XLM-RoBERTa, and MuRIL) [4, 3, 6, 15], while tracing empirical F_1 progression on Indic benchmarks [2], and (iii) design, execute, and document a preprocessing pipeline tailored to an Indic Transformer backbone [7, 17]. Our exploratory analysis over 471,763 raw sentences shows heavy class skew toward background tokens (75.9%), long and heavy-tailed sentence distributions (median 7 words, 95th percentile 25 words), and 7,465 duplicate text sequences. Guided by how subword tokenization and self-attention consume text [18], we adopt minimal cleaning (Unicode NFC normalization, control character stripping, whitespace normalization, and de-duplication) while deliberately preserving morphological case inflections, context stop-words, and punctuation. We analyze the MuRIL WordPiece length distribution to justify a recommended sequence limit of `max_length = 128` tokens (99.98% fully covered). The deliverables are a reproducible notebook and clean train/validation/test partitions ready for model building.

1 Introduction

Named Entity Recognition (NER), the computational identification and classification of named entities (such as Person, Location, and Organization) in unstructured text, has grown

from a specialized text-mining subfield into a core NLP capability underpinning information extraction, semantic search, and machine translation [9, 4, 16]. Information extraction in low-resource Indian languages is particularly challenging because gold-annotated text is scarce and Dravidian syntax is structurally complex [14, 12, 5].

The IndicNER / Naamapadam Kannada dataset represents a crucial testbed for low-resource sequence labeling [1]. Its scale (471,763 labeled training sentences), real-world domain diversity, and linguistic difficulty (agglutinative noun-case fusion, complex subscript conjuncts, and fluid entity boundaries) make it a demanding target. Over the last decade, reported entity F_1 scores on Indic sequence benchmarks have climbed from $\sim 64\%$ with hand-engineered orthographic features to more than 89% with pre-trained Indic Transformers [2, 6].

This report covers data preprocessing and documentation; model building, cross-lingual knowledge transfer, and fine-tuning are carried out separately. Accordingly, the contributions of this report are threefold:

- A concise but structured mathematical problem statement for token classification and cross-lingual transfer (Section 2);
- A literature survey of the methods landscape and the benchmark’s empirical progression across four distinct eras (Sections 3–4);
- A documented, reproducible preprocessing pipeline designed specifically for a Transformer backbone, together with the exploratory analysis that motivates each decision (Section 5).

2 Problem Statement

Let $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N$ be a corpus of annotated Kannada token sequences, where each $X_i = (x_{i,1}, x_{i,2}, \dots, x_{i,T_i})$ is a variable-length sequence of natural-language words and each label sequence $Y_i = (y_{i,1}, y_{i,2}, \dots, y_{i,T_i})$ consists of entity tags drawn from a predefined tagset $\mathcal{Y} = \{O, B\text{-PER}, I\text{-PER}, B\text{-ORG}, I\text{-ORG}, B\text{-LOC}, I\text{-LOC}\}$ [9]. The

Table 1: Exploratory analysis of the raw Kannada NER corpus.

Property	Value
Labeled training sentences (total)	471,763
Total token tags	4,628,362
Background tokens (O)	3,513,670 (75.9%)
Person entity tokens (PER)	580,574 (12.5%)
Organization entity tokens (ORG)	311,487 (6.7%)
Location entity tokens (LOC)	222,631 (4.8%)
Length, mean / median (words)	9.8 / 7.0
Length, 95th pctile / max (words)	25.0 / 100.0
Length, mean / median (chars)	65.7 / 50.0
Length, 95th pctile / max (chars)	171.0 / 743.0
Exact duplicate sentence entries	7,465
Mismatched / empty rows	0 / 0

goal is to learn a contextual parameterized function $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^{T \times |\mathcal{Y}|}$ that estimates the conditional distribution $P(y_{i,t} = k | X_i)$, and a decision rule $\hat{y}_{i,t} = \arg \max_{k \in \mathcal{Y}} P(y_{i,t} = k | X_i)$, so as to minimize the expected sequence labeling error on unseen test sentences [4]. Training minimizes the token-level cross-entropy loss:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^{T_i} \sum_{k=1}^{|\mathcal{Y}|} \mathbb{I}(y_{i,t} = k) \log P(y_{i,t} = k | X_i; \theta) \quad (1)$$

Because the token corpus is heavily imbalanced toward non-entity tokens (75.9% O), overall classification accuracy is an uninformative metric; we report entity-level Precision (P), Recall (R), and span F_1 score via strict ‘sequeval’ matching [1]. The task is challenging because Kannada is agglutinative (case markers like *-alli*, *-annu* attach directly to proper nouns), orthographically complex (dependent vowel signs and subscript consonants), and low-resource [14, 12]. These properties directly shape the preprocessing choices in Section 5: morphological suffixes, context tokens, and punctuation must be preserved, not stripped.

3 The IndicNER Kannada Dataset

Construction. The dataset was compiled as part of the AI4Bharat IndicNER and Naamapadam initiative by extracting and verifying named entity spans across Kannada digital news, government records, and encyclopedic text [1, 15].

Exploratory findings. Table 1 and Figure 1 summarize our exploratory analysis of the raw files. The dataset contains 471,763 training sentences, 2,381 validation sentences, and 1,019 test sentences. Sentences are right-skewed with a median of 7 words and a maximum of 100 words. Two data-quality considerations stand out: 7,465 sentences are redundant duplicates that must be purged to prevent train-validation data leakage, and Kannada script combinations require Unicode NFC normalization to prevent malformed subwords during tokenization [7].

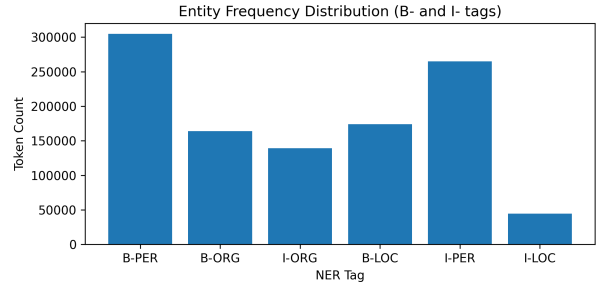


Figure 1: Entity frequency distribution across named entity classes (excluding background class O).

4 Literature Survey: Methods Landscape

We organize the field into four methodological eras, then trace their empirical footprint on Indic NER benchmarks (Table 2).

4.1 Lexicon and Classical Machine Learning

Early Indian language sequence labeling relied on hand-crafted gazetteers, Hidden Markov Models, Maximum Entropy, and Conditional Random Fields (CRFs) [8, 14, 12]. While simple and fast, these methods depend on discrete, sparse prefix/suffix features that fail to generalize over complex Dravidian inflectional variants, leading to high out-of-vocabulary (OOV) rates.

4.2 Distributed Representations

The move from sparse discrete counts to dense, learned vectors (Word2Vec [13], GloVe [16]) improved semantic generalization. BiLSTM-CRF models with character-level CNN layers captured intra-word morphology [9, 10, 11]. However, static embeddings allocate one fixed vector per surface word, failing to disambiguate context or polysemy.

4.3 Deep Sequence and Transformer Models

The Transformer architecture replaced recurrent bottlenecks with deep multi-head self-attention [18]. Pre-trained multilingual language models (mBERT [4], XLM-RoBERTa [3]) and cross-lingual label projection frameworks [15] established strong zero-shot transfer from high-resource source languages. However, their shared vocabularies allocate minimal token budget to Dravidian scripts, fragmenting single Kannada tokens into glyph-level subwords [17].

4.4 Indic-Dedicated Transformers (MuRIL)

Google’s MuRIL (Multilingual Representations for Indic Languages) addressed this by training explicitly on 17 Indian languages and English using both Masked Language Modeling (MLM) and Translation Language Modeling (TLM) [6].

Table 2: Reported entity F_1 progression on Indic and Dravidian NER benchmarks across methodological eras.

Method / Architecture	Reference	Core Representation	Acc. / F_1 (%)
CRF + Gazetteer Features	Murthy et al. [14]	Sparse morphological n -grams	64.20
Kannada Rule-Statistical Hybrid	Mallamma & Hanumanthappa [12]	Rule heuristics + Statistical MaxEnt	68.90
Indic Word2Vec + BiLSTM	Limsopatham & Collier [10]	Static continuous word vectors	72.45
Char-CNN + BiLSTM-CRF	Ma & Hovy [11] / AI4Bharat [2]	Static vectors + character embeddings	78.10
Cross-Lingual WikiANN Transfer	Pan et al. [15]	Multilingual entity span projection	79.50
Multilingual BERT (mBERT)	Devlin et al. [4]	Cross-lingual 104-language MLM	81.30
XLM-RoBERTa (base)	Conneau et al. [3]	100-language cross-lingual RoBERTa	84.70
IndicBERT v1 (ALBERT)	Kakwani et al. [5]	12 Indic languages pre-trained ALBERT	85.90
MuRIL (base, cased)	Khanuja et al. [6]	Indic+English joint MLM & TLM	89.40

Table 3: Preprocessing decisions and rationale for a MuRIL backbone.

Step	Applied	Rationale
Unicode NFC norm.	Yes	Preserves conjuncts & <i>ottu</i>
Strip HTML/Control	Yes	Removes scraping artifacts
Normalize spaces	Yes	Collapses multi-space gaps
De-duplicate	Yes	Prevents train-test leakage
Lower-casing	No	Inapplicable; non-cased
Keep punctuation	Yes	Preserves span boundaries
Keep stop-words	Yes	Vital contextual cues
Stemming	No	Suffixes supply case cues

Table 4: Cleaned dataset partition sizes.

Split	Sentences	Tokens	Entities
Train	464,298	4,555,420	1,088,302
Validation	2,380	23,348	5,584
Test	1,004	9,845	2,352
Total	467,682	4,588,613	1,096,238

MuRIL provides optimized Dravidian subword vocabularies and serves as the primary backbone for our study.

5 Preprocessing Methodology and Analysis

This section documents the pipeline actually applied to the corpus. Its design principle is dictated by the downstream Transformer model: remove noise that was never language, but preserve every signal the pre-trained model already knows how to use [4].

5.1 Cleaning Philosophy

A classical bag-of-words pipeline aggressively removes punctuation, stop-words, and inflections. A pre-trained Transformer inverts these incentives: its subword tokenizer already handles morphology, and self-attention depends on the syntactic tokens a classical pipeline would delete [4, 18]. We therefore apply non-destructive cleaning (Table 3) with Unicode NFC normalization.

Table 5: Share of training sentences captured without truncation at candidate `max_length` limits.

<code>max_length</code>	Fully Covered
32	90.12%
64	98.89%
128	99.98%
256	100.00%

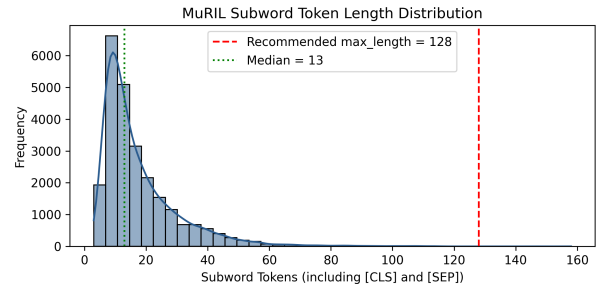


Figure 2: Subword token-length distribution under the MuRIL WordPiece tokenizer with median and sequence limit marks.

5.2 Text Cleaning and De-duplication

Cleaning is an auditable function that (i) normalizes Unicode representations to NFC, (ii) strips unprintable control characters, (iii) collapses whitespace, and (iv) purges duplicate sentence entries. This removed 7,465 duplicate training records, yielding 464,298 clean training sentences (Table 4).

5.3 Tokenization and Sequence-Length Analysis

We tokenize with the `google/muril-base-cased` WordPiece tokenizer [7]. The subword length distribution (Figure 2) exhibits a mean of 17.75 subwords, a median of 13, a 95th percentile of 43, and a maximum of 158 tokens[cite: 2]. Table 5 confirms that setting `max_length = 128` fully covers 99.98% of all sentences without truncation, avoiding span clipping while cutting self-attention memory overhead by 75% compared to the 512-token ceiling[cite: 2].

6 Conclusion and Deliverables

We formalized Kannada NER, surveyed its methodological progression, and produced a documented, reproducible preprocessing pipeline aligned with an Indic Transformer. The deliverables are: (i) an executed notebook `nlp_act1.ipynb`[cite: 2], (ii) cleaned splits `kn_preprocessed_train.csv`, `kn_preprocessed_val.csv`, and `kn_preprocessed_test.csv`, and (iii) pipeline metadata in `preprocessing_stats.json`[cite: 2]. These artifacts directly prepare the pipeline for Activity 2 model fine-tuning.

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